

Alex McAnnally

Highly skilled data professional with over a decade of experience in database administration, data migration, and systems management. Adept at designing and optimizing secure, scalable solutions in SaaS and healthcare IT environments, with proven success in managing sensitive datasets like Medicare and Medicaid records. Skilled in translating complex technical concepts into actionable strategies for stakeholders of all technical backgrounds. Expertise includes SQL (T-SQL, MySQL, PostgreSQL), Python, PowerShell, Docker, and advanced data analysis tools.

I have also contributed to academic and clinical research initiatives through advanced data analytics and technical consulting, enabling data-driven discoveries and evidence-based decision-making. Committed to delivering innovative solutions that drive efficiency, maintain compliance in regulated industries, and support meaningful research outcomes.

EXPERIENCE

UAB School of Medicine, Birmingham, AL

March 2022 – Present

Database Analyst III

- Directs design of new databases/files and fine-tune the performance of existing Microsoft Windows and Linux file servers.
- Responsible for managing the file servers, network infrastructure, and large Medicare, Medicaid, and MarketScan databases for the PEER group (120+ users).
- Consults with members of the department as required to perform problem determination.
- Maintains database recoverability.
- Provides support to the department as a technical consultant for planning and implementing systems with ready accessibility to appropriate data.
- Manages the electronic transfer and protection of confidential information from corporate sponsors and other agencies external to UAB in a variety of data formats.
- Controls the security of all data in the database.
- Provides security training and technical guidance for users who access the PEER network.
- Handles special file-related projects.
- Provides technical and administrative support to the department.

CONTACT

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EDUCATION

Western Governor's University,
Salt Lake City, UT (online)

2016 – 2018

Bachelor of Science in
Information Technology

**University of Alabama at
Birmingham, Birmingham, AL**

2024 – 2025

Master of Science in Management
Information Technology –
Business Analytics

TECHNOLOGY SKILLS

Data Engineering / Database
Tools:

- ETL/ELT Pipelines (design and optimization)
- SQL Server Management Studio (SSMS)
- SQL Server Agent / Job Scheduling
- BCP / SQLCMD utilities
- Oracle SQL Developer
- Query Optimization & Indexing
- Database Performance Tuning
- Database Security & Compliance (PHI/PII, CMS data)

Analytics / Data Science:

- Pandas / NumPy
- NLTK / Text Analytics
- scikit-learn (classification, regression, model evaluation)
- KNIME Analytics Platform
- Tableau (dashboards, KPIs, calculated fields, LODs)
- CRISP-DM Framework

Version Control & DevOps:

- CI/CD Basics (GitHub Actions)
- PowerShell Automation & Scripting

Programming Languages & Scripting:

- Python (data processing, automation, machine learning)
- JavaScript (general scripting)
- VB.NET
- R (tidyverse, modeling)
- PowerShell scripting
- Bash scripting
- SAS (data manipulation, macros, SPDE libraries)

UAB School of Medicine, Birmingham, AL

August 2018 – March 2022

Database Analyst II

- Provide technical support to programming staff in database programming, design, and maintaining data files through regular backups.
- Responsible for management of very large data sets of clinical, genetic, proteomic data from human subjects in clinical research studies.
- Perform database administration by maintaining data files and integrity of the database through backup, and problem determination.
- Technical consultant to faculty members developing research proposals.
- Assist with system efficiency by monitoring file placement, reorganization, etc.
- Complete extensive programming for data manipulation and to allow biostatisticians to perform detailed statistical analysis on very large data sets.

Hyland Software, Cleveland, OH (remote)

May 2016 – August 2018

Database Engineer II

- Work with key customer stakeholders to gather requirements and develop a plan for technical delivery
- Perform technical analysis on customer databases and gathering data to implement proper database solution
- Document specifications or findings and recommendations resulting from database consulting services in a manner consistent with Hyland Software documentation standards
- Implement database solutions for customers; activities may include general database consulting, executing miscellaneous scripting requests against customer database, performing database migrations and upgrades, legacy conversions or building custom reports.
- Maintain proficiency in third party and Hyland specific technology necessary to successfully implement database solutions
- Collaborate with peers and contribute to existing Knowledge Base regarding technical information gathered while performing services

Daxko, Birmingham, AL

May 2014 – April 2016

Data Conversion Engineer

- Effectively communicate with clients to obtain, map and validate all aspects of the data conversion.
- Script and build client sites using developed tools.
- Manipulate client data using SQL, T-SQL, Excel, or custom tools developed in Visual Studio (C#, ASP.NET) to the format required for Daxko's products.
- Develop and deploy mobile applications using Android Studio and XCode.
- Load data into clients' data validation and training databases.
- Validate data to ensure that it is consistent, well formed, and of high structural integrity.
- Adhere to coding standards and help in process documentation to ensure efficiency.
- Report on data integrity issues to Implementation team members and clients.
- Work with multiple stakeholders (e.g. project managers, engineers, clients) and develop effective solutions to complex problems.

CERTIFICATIONS

- AWS Certified Cloud Practitioner
- Microsoft Certified Technology Specialist: Windows 7 (MCTS)
- Microsoft Certified Professional (MCP)
- CompTIA Linux+
- CompTIA Project+
- CompTIA Security+
- CompTIA A+

PUBLICATIONS

"A Markov Chain Replacement Strategy for Surrogate Identifiers: Minimizing Re-Identification Risk While Preserving Text Reuse", Electronics (MDPI), October 2025

"Classifying Multimorbidity Using Drug Concepts via the RxRisk Comorbidity Index", University of Alabama at Birmingham, Division of Immunology and Rheumatology, November 2022

"Influence of Multimorbidity on New Treatment Initiation and Achieving Target Disease Activity Thresholds in Active Rheumatoid Arthritis: A Cohort Study Using the Rheumatology Informatics System for Effectiveness Registry", Arthritis Care & Research, August 2021

"Mapping Multimorbidity Using Drug Concept Unique Identifiers (RxCUIs) via the Rx-Risk Comorbidity Index", University of Alabama at Birmingham, Division of Immunology and Rheumatology, November 2020